



NYU

Information
Technology

Volatility Analysis (AAPL)

Quantitative Methods in Finance

Group 4

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Overview

1. Implied Volatility
2. Historical Volatility
3. VRP and Long Straddle Trading Strategy
4. Extension to a time-series strategy, backtest, and Performance Metrics

AAPL Implied Volatility

$$C_{market} = SN(d_1) - Ke^{-rT}N(d_2)$$

$S = 280.14$
 $T = 0.0767$
 $r = 0.0372$

Apple Inc. (AAPL) ☆

280.14 **+8.79 (+3.24%)** **280.07** **-0.07 (-0.02%)**

At close: May 1 at 4:00:02 PM EDT

After hours: May 1 at 7:59:59 PM EDT

Trade Apple on Coinbase

Trading disclosure ⓘ

May 29, 2026 ▾

All Strike Prices ▾

List ▾

All Options ▾

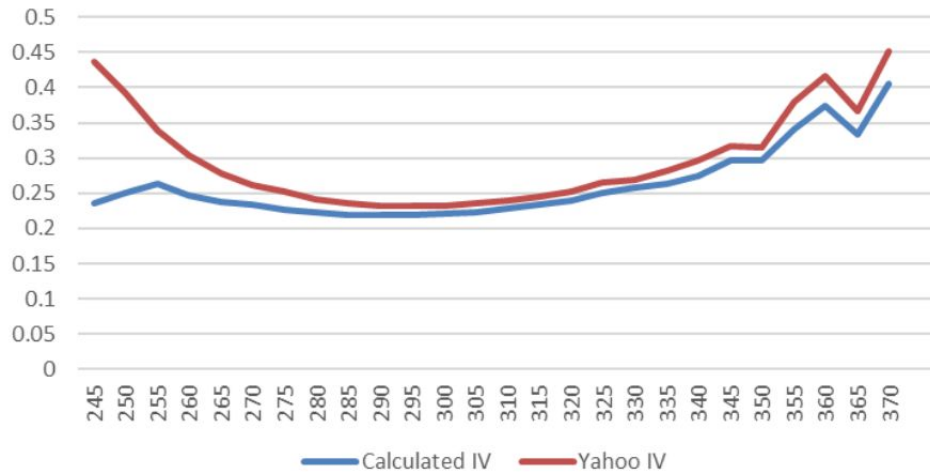
Calls

In The Money

Contract Name	Last Trade Date (EDT)	Strike	Last Price	Bid	Ask	Change	% Change	Volume	Open Interest
AAPL260529C00110000	5/1/2026 3:08 PM	110	170.83	168.20	172.05	+8.83	+5.45%	1	1
AAPL260529C00195000	4/24/2026 10:27 AM	195	77.74	83.60	87.35	+0.00	+0.00%	1	1
AAPL260529C00200000	4/23/2026 3:27 PM	200	74.86	78.55	82.20	+0.00	+0.00%	2	79
AAPL260529C00205000	4/17/2026 1:42 PM	205	66.48	73.75	77.45	+0.00	+0.00%	6	26
AAPL260529C00210000	4/17/2026 10:12 AM	210	59.89	68.60	72.55	+0.00	+0.00%	4	3
AAPL260529C00215000	5/1/2026 11:11 AM	215	67.76	63.70	67.50	+8.88	+15.08%	3	3
AAPL260529C00220000	5/1/2026 11:11 AM	220	62.85	58.75	62.65	+14.37	+29.64%	3	33
AAPL260529C00225000	4/23/2026 1:56 PM	225	48.93	54.20	56.85	+0.00	+0.00%	2	4
AAPL260529C00230000	5/1/2026 9:48 AM	230	53.70	49.00	51.95	+15.92	+42.14%	10	15
AAPL260529C00235000	4/30/2026 3:06 PM	235	39.91	44.40	47.00	+0.00	+0.00%	26	70
AAPL260529C00240000	4/30/2026 3:59 PM	240	34.00	39.45	42.10	+0.00	+0.00%	23	59
AAPL260529C00245000	5/1/2026 10:16 AM	245	42.36	34.65	37.25	+12.92	+43.89%	4	33
AAPL260529C00250000	5/1/2026 3:02 PM	250	31.80	30.10	32.30	+6.35	+24.95%	42	361

strike	mid_price	IV_calc	BS_price_from_yahoo_IV	impliedVolatility
250	31.200	0.250110	32.921331	0.392096
255	26.700	0.263289	27.819056	0.338386
260	22.075	0.247517	23.063749	0.303352
265	17.750	0.238394	18.600199	0.277961
270	13.825	0.233067	14.575289	0.262336
275	10.300	0.226633	11.034078	0.251900
280	7.350	0.222377	7.923619	0.240974
285	5.000	0.219233	5.500763	0.235725
290	3.275	0.218385	3.671412	0.232735
295	2.070	0.218829	2.396376	0.232796
300	1.275	0.220529	1.490376	0.232185
305	0.775	0.223453	0.937735	0.235115
310	0.485	0.228968	0.601305	0.240242
315	0.300	0.234142	0.383326	0.245369
320	0.190	0.240319	0.254934	0.252449
325	0.135	0.250431	0.196100	0.265144
330	0.090	0.257545	0.125392	0.269539
335	0.060	0.264226	0.100810	0.282234
340	0.045	0.274122	0.087325	0.296882
345	0.050	0.295917	0.087490	0.316413
350	0.030	0.297143	0.050688	0.314460
355	0.065	0.341693	0.148311	0.378912
360	0.090	0.373516	0.205944	0.416510
365	0.020	0.333479	0.051613	0.367194
370	0.085	0.406158	0.194059	0.451177

Yahoo IV vs. Calculated IV



- Calculated IV closely tracks Yahoo IV across most strike prices
- Both curves exhibit a clear volatility smile pattern
- Minimum IV occurs near at-the-money region
- Differences are larger for deep in-the-money options
- Discrepancies decrease as strike approaches ATM

AAPL Historic Volatility

Methods

Rolling Window

$$\sigma_t = \left(\frac{1}{n-1} \sum_{i=0}^{n-1} (r_{t-i} - \bar{r})^2 \right)^{1/2}$$

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

$$\sigma_{\text{annual}}^2 = 252 \times \sigma_{\text{daily}}^2$$

EWMA

$$\sigma_t^2 = \lambda \cdot \sigma_{t-1}^2 + (1 - \lambda) \cdot r_{t-1}^2$$

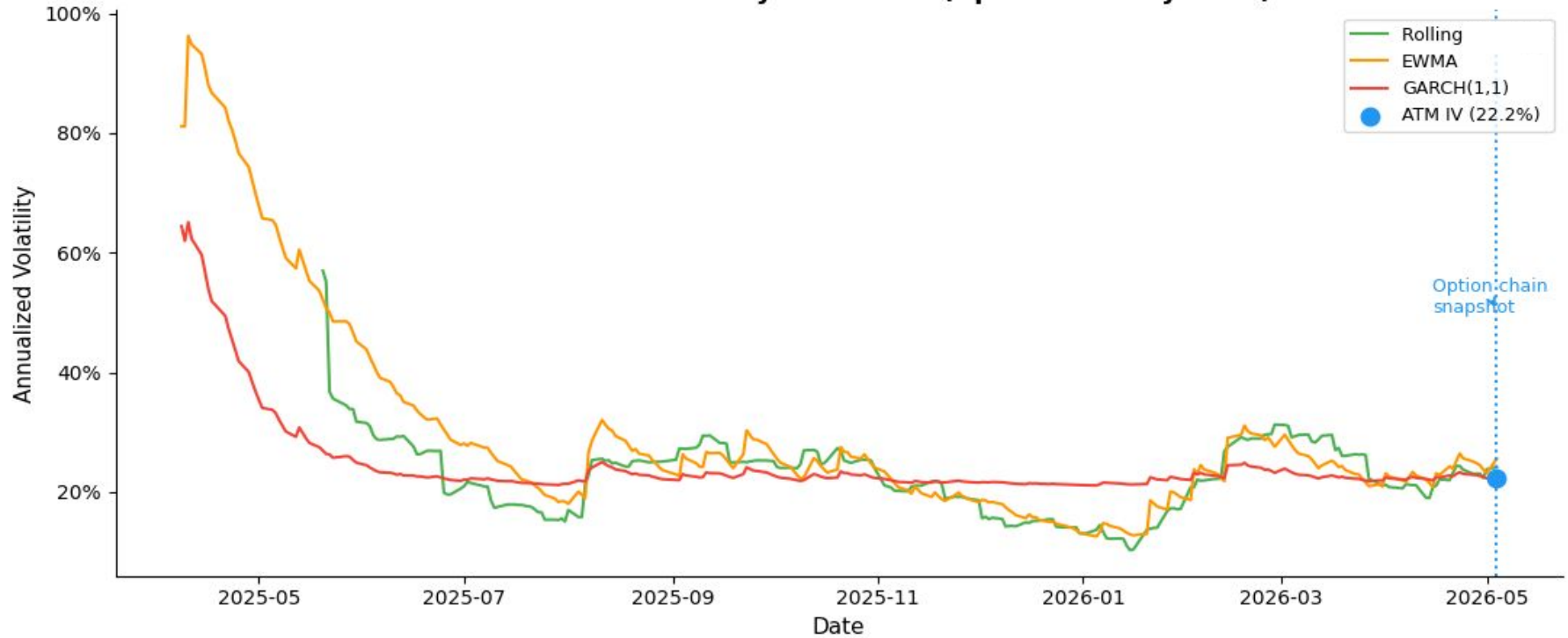
$$\sigma_t^2 = (1 - \lambda) \sum_{i=1}^{\infty} \lambda^{i-1} \cdot r_{t-i}^2$$

GARCH(1,1)

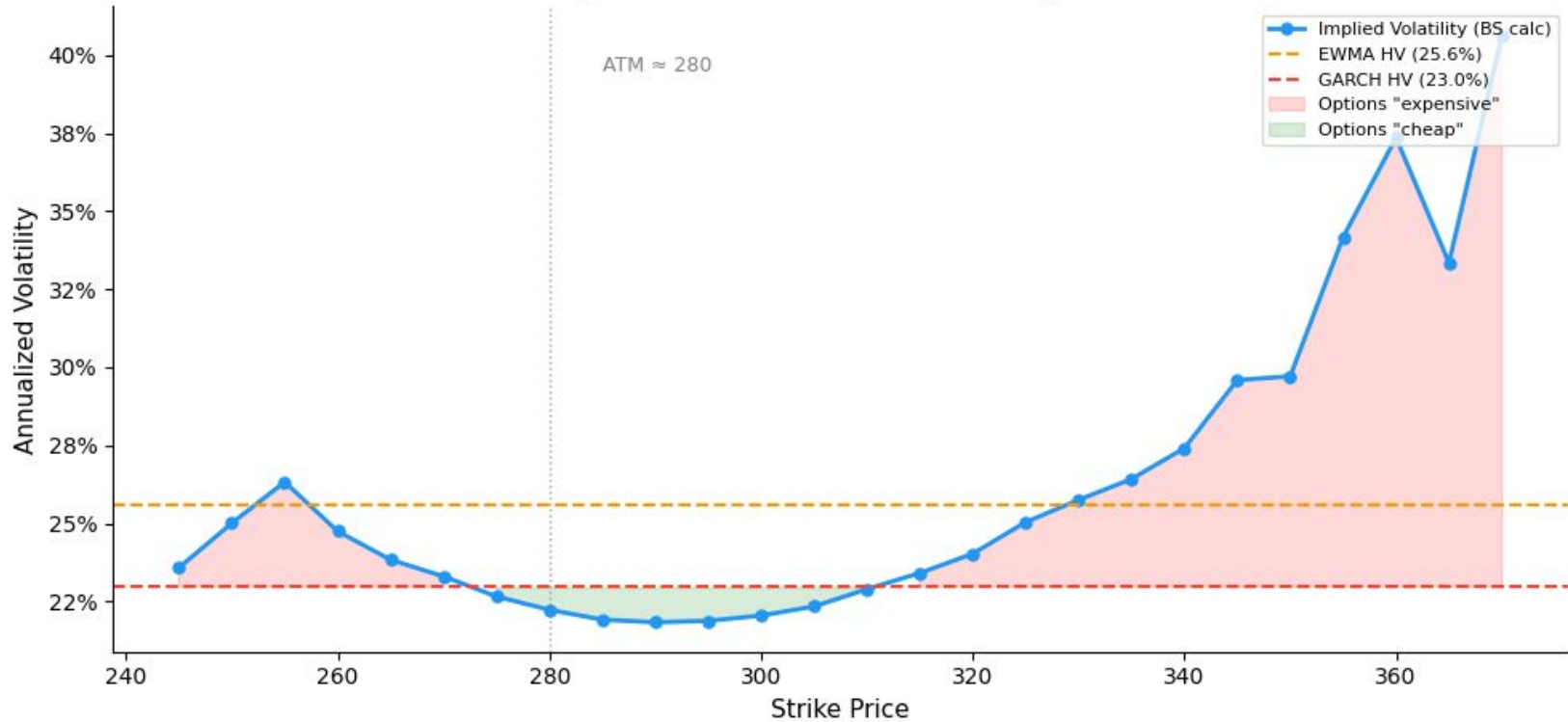
$$\sigma_t^2 = \omega + \alpha \cdot r_{t-1}^2 + \beta \cdot \sigma_{t-1}^2$$

$$\bar{\sigma}^2 = \frac{\omega}{1 - \alpha - \beta}$$

AAPL Historical Volatility Over Time (Apr 2025 - May 2026)



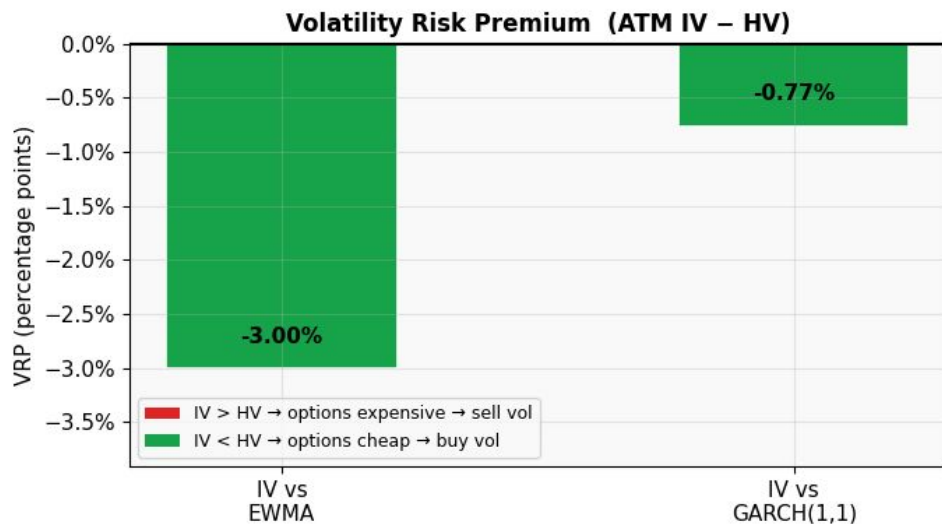
Volatility Smile vs Historical Volatility Estimates



AAPL Trading Strategies

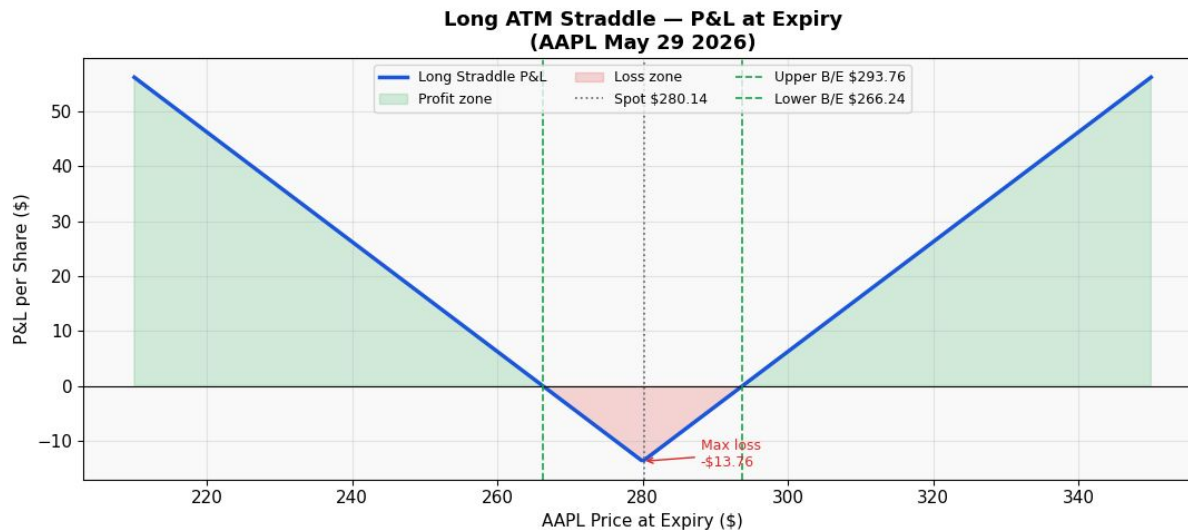
Strategy - 1

Volatility Risk Premium (VRP)



- $VRP = \sigma_{IV} - \sigma_{HV}$
- **VRP < 0** : IV is cheap → buy vol
 - VRP : EWMA HV = **-3.00%**
 - VRP : GARCH HV = **-0.77%**
- Market IV is lower than realized/model volatility
- Options are relatively cheap vs historical movement
- Market is underpricing volatility price
- **IMPLICATION:**
 - This creates a long volatility opportunity

Long Straddle Payoff Profile



Premium: \$13.76

- Call: \$7.35
- Put: \$6.41

- Profit only if large move happens
- Loss capped at premium (\$13.76)

Breakeven:

- Upper: \$293.76
- Lower: \$266.24

Trade: Long ATM Straddle

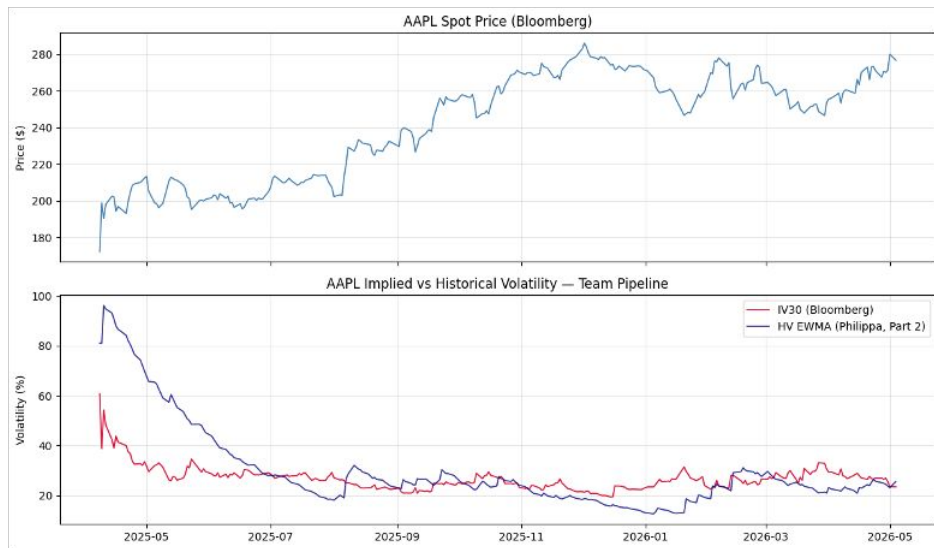
- Long ATM straddle constructed by **buying both call and put at \$280 strike**.
- Position profits only from **large price movement**, independent of direction.
- Profitability requires price movement beyond breakeven levels of **\$266.24** or **\$293.76**.
- Maximum loss is limited to premium paid, equal to **\$13.76** in this structure.
- If price remains **near strike**, both options **decay toward zero** at expiry.

Formulas

- Premium = $C + P$
- Call payoff at expiry: $CT = \max(S_T - K, 0)$
- Put payoff at expiry: $PT = \max(K - S_T, 0)$
- Long straddle payoff = $\max(S_T - K, 0) + \max(K - S_T, 0) - \text{Premium}$
- Breakevens
 - $BE_{\text{upper}} = K + \text{Premium}$
 - $BE_{\text{lower}} = K - \text{Premium}$
- Trade Condition: $VRP < 0 \Rightarrow \text{Long Straddle}$
- Put-call parity: $P = C - S + K e^{-rT}$

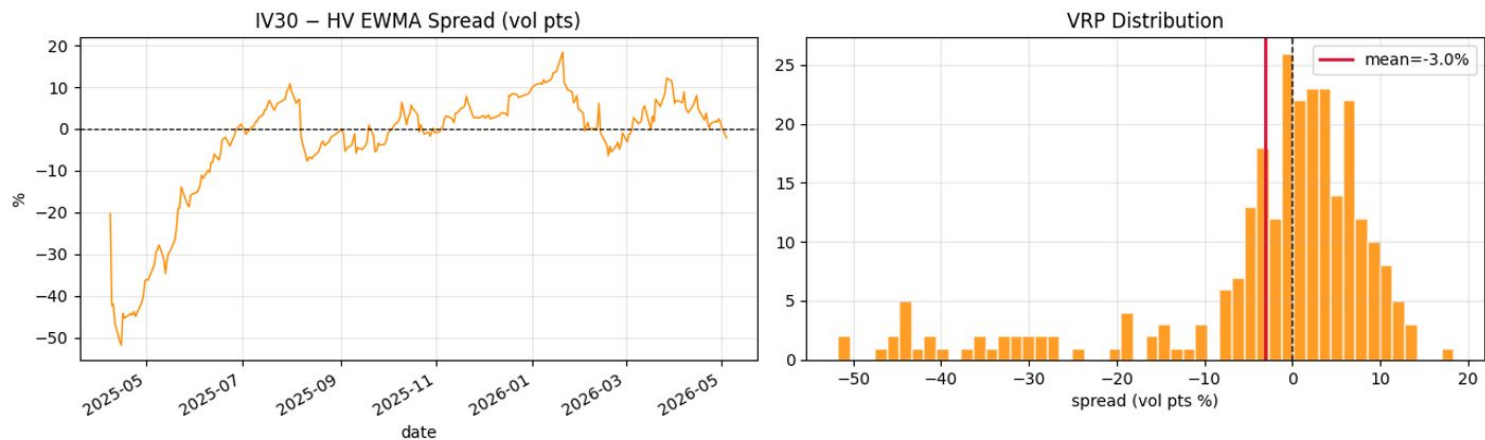
Strategy - 2

IV30 vs HV EWMA



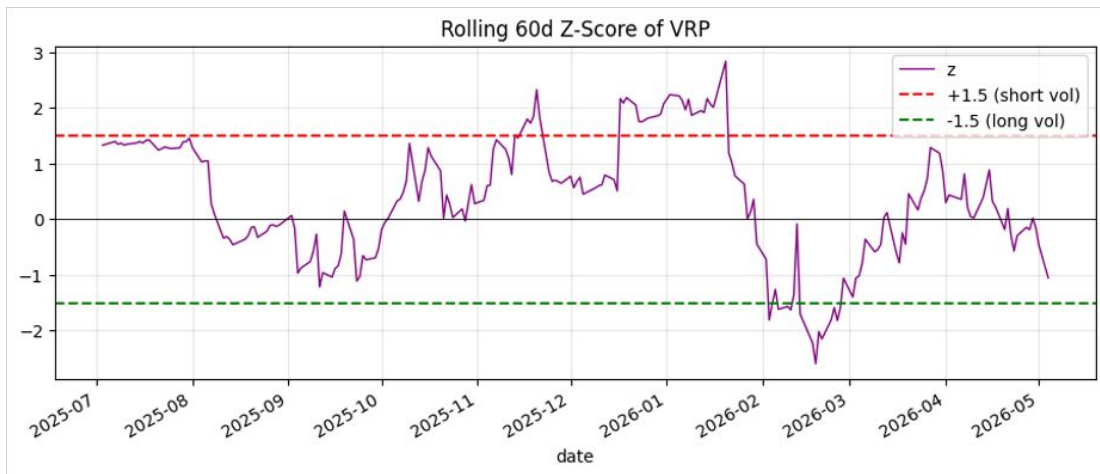
- Rolling HV vs Bloomberg HV30: correlation 0.97, MAE 0.79 vol pts
- April 2025 tariff shock: HV spiked above IV $\rightarrow$ short-vol regime breaks
- A static rule fails. We need a signal that flips with the regime.

The VRP



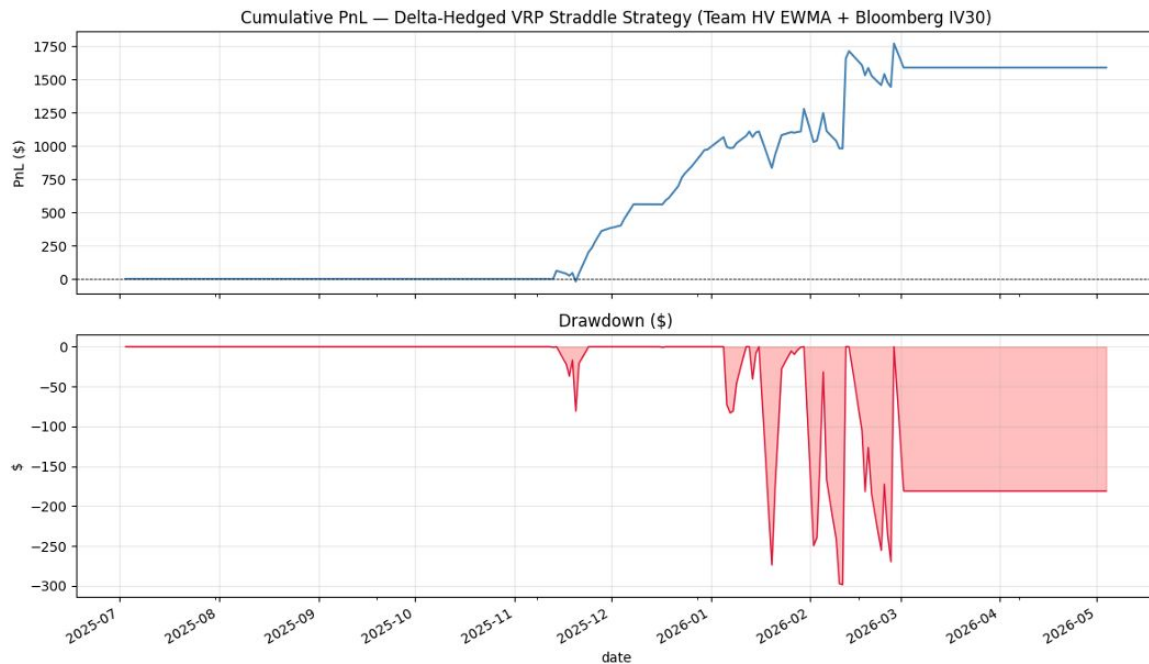
Mean: -3.0 vp **Median:** +0.6 vp Days >0: 55% **5th pctile:** -38.6 vp **95th pctile:** +10.8 vp

Trading Signal



Signal	Z - Score	Action
Short V.	$Z > 1.5$	Sell Straddle
Long V.	$Z < -1.5$	Buy Straddle
Flat	$ Z < 1.5$	No Action

Backtest



Metric	Value
Sharpe	1.76
PnL	\$1,588
Max Drawdown	-\$299
Hit Rate	67.2%
Win/Loss (days)	45/22
Win/Loss (Ratio)	0.97
Avg Win Day	\$71
Avg Loss Day	-\$73
In-position	31%

Conclusion

- VRP has negative mean, positive median. April 2025 flips the sign of the average. Static long-vol or short-vol bets would perform worse.
- Rolling 60-day standardisation, threshold $|z| > 1.5$. In position 20% of days.
- Walk-forward validation, real option quotes, earnings blackout, and realistic transaction costs are the natural next steps before any live capital.

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Thank You



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